

Stance on a claim over time (experimental)

```
st <- list(
  name = "world_role", # short name for this claim's output files
  proposition = "The United States should take an active role in the affairs of
other nations.",
  negation = "The United States should stay out of the affairs of other nations
and avoid foreign commitments.",
  strata = c(high = 0.05, mid = 0.20), # top 5% by similarity, then the next
20%; the rest is low
  n = c(high = 200L, mid = 70L, low = 30L), # calls per stratum; unused calls
roll down
  seed = 2026L,
  window_n = 300L, # calls per window
  window_wait = 60L, # seconds between windows; 4 * 3600 for a rate-limited
large model
  df_spline = 2L, # flexibility of the trend
  min_class_n = 5L, # paragraphs needed on each side for any trend
  min_year = NULL, # optional: fit trends from this year on
  use_openrouter = TRUE, # FALSE at work
  steps = tibble::tribble(
    ~step, ~base_url_env, ~api_key_env, ~model,
    "embed", "COMPASS_EMBED_URL", "COMPASS_EMBED_KEY", "MODEL-FOR-EMBEDDING",
    "stance", "COMPASS_SMALL_URL", "COMPASS_SMALL_KEY", "SMALL-MODEL"))

suppressPackageStartupMessages({
  library(here); library(splines); library(viridis)
  library(tidyverse) # tidyverse last so its verbs win
})
theme_set(theme_minimal(base_size = 12))
source(here::here("code", "llm_source.R"))
if (isTRUE(st$use_openrouter)) {
  source(here::here("code", "llm_openrouter.R"))
  st$steps$model <- open_router_models[st$steps$step]
}
```

The design

Coding every paragraph would cost one model call each. Instead, every paragraph is scored for free by how close its meaning is to the claim, and a sample is coded: heavier where the claim is likely to appear, never zero anywhere. This is two-phase stratified sampling. Similarity is the first-phase measure and the model's label the second, and each coded paragraph carries the weight N/n of its stratum, so estimates describe the whole corpus rather than only the paragraphs that were read.

1. **Similarity.** The claim and its negation are embedded with the same model as the corpus (2 embedding calls). Each paragraph gets the higher of its two cosines, so paragraphs that argue against the claim in their own words rank as high as ones that echo it.
2. **Strata.** Paragraphs are ranked by similarity and split into high, middle, and low strata.
3. **Coding.** A sample from each stratum is labeled favor, neutral, oppose, or irrelevant, one call per paragraph. The low stratum doubles as the audit of what the similarity screen misses.

Claim: The United States should take an active role in the affairs of other nations.

Negation used for scoring: The United States should stay out of the affairs of other nations and avoid foreign commitments.

```
para_in <- p_out(c("paragraphs.csv", "paragraph_vectors.rds"))
paras <- readr::read_csv(para_in[1], show_col_types = FALSE)
V <- readRDS(para_in[2])[paras$para_uid, , drop = FALSE]
q <- embed_texts(c(st$proposition, st$negation), st$steps, st$use_openrouter)
if (ncol(q) != ncol(V))
  stop("The claim was embedded with a different model than the corpus. ",
       "Use the same embed row as preprocess.qmd.")
unit <- function(m) m / sqrt(rowSums(m^2))
sims <- unit(V) %*% t(unit(q))
paras <- paras |>
  dplyr::mutate(sim = pmax(sims[, 1], sims[, 2]),
               pct = dplyr::percent_rank(dplyr::desc(sim)),
               stratum = dplyr::case_when(
                 pct < st$strata[["high"]] ~ "high",
                 pct < st$strata[["high"]] + st$strata[["mid"]] ~ "mid",
                 TRUE ~ "low"))
writeLines(glue::glue("{nrow(paras)} paragraphs from \\
  {dplyr::n_distinct(paras$atom_id)} comments scored against the claim."))
```

951 paragraphs from 60 comments scored against the claim.

```
# A stratum smaller than its allocation is coded in full and its unused
# calls roll down to the next stratum (high, then mid, then low), so the
# whole budget is spent.
N <- table(factor(paras$stratum, levels = c("high", "mid", "low")))
alloc <- purrr::reduce(c("high", "mid", "low"), function(acc, h) {
  want <- st$n[[h]] + acc$spare
  take <- min(N[[h]], want)
  list(n = c(acc$n, stats::setNames(take, h)), spare = want - take)
}, .init = list(n = integer(), spare = 0))$n
sampled <- paras |>
  dplyr::group_split(stratum) |>
  purrr::map(function(d) {
    n_h <- alloc[[d$stratum[1]]]
    i <- withr::with_seed(st$seed, sample.int(nrow(d), n_h))
    dplyr::mutate(d[sort(i), ], N_h = nrow(d), n_h = n_h, w = N_h / n_h)
```

```

}) |>
purrr::list_rbind()
sampled |>
  dplyr::group_by(stratum) |>
  dplyr::summarise(paragraphs = dplyr::first(N_h), coded = dplyr::first(n_h),
                  weight = round(dplyr::first(w), 1)) |>
  dplyr::left_join(paras |>
                  dplyr::group_by(stratum) |>
                  dplyr::summarise(similarity = str_c(round(min(sim), 2), "
to ",
                                                    round(max(sim), 2))),
                  by = "stratum") |>
  dplyr::arrange(match(stratum, c("high", "mid", "low"))) |>
  dplyr::select(stratum, similarity, paragraphs, coded, weight) |>
  tbl_paged(caption = "Strata. Each coded paragraph stands for `weight` paragraphs
of its stratum.")

```

Strata. Each coded paragraph stands for weight paragraphs of its stratum.

stratum	similarity	paragraphs	coded	weight
high	0.84 to 0.87	48	48	1.0
mid	0.81 to 0.84	190	190	1.0
low	0.72 to 0.81	713	62	11.5

Coding

One paragraph per call, at temperature 0, without the year. Labels are saved per paragraph after every window of 300 calls, so an interrupted run resumes where it stopped, and a changed sample only sends paragraphs not yet coded. Changing the claim's wording refuses the saved labels and names the file to delete.

```

labels_path <- p_out(str_c("stance_labels_", st$name, ".csv"))
prompt_path <- p_out(str_c("stance_prompt_", st$name, ".txt"))
sys <- build_prompt(
  role = "You are a survey methodologist coding stance in public comments.",
  task = str_c("Decide the author's position toward this proposition: \"",
st$proposition,
              "\". First decide whether the text addresses the proposition at
all; ",
              "if not, the label is irrelevant. Otherwise decide favor, oppose,
or neutral."),
  rules = c("Stance is measured only toward the proposition.",
            "Stance is not tone: an angry passage can favor the proposition.",
            "Judge only the text shown; use no outside knowledge.",
            "If the text is ambiguous, choose neutral rather than guessing."),

```

```

    output = "One label: favor, neutral, oppose, or irrelevant.")
writeLines(sys, prompt_path)
check_inputs(labels_path, c(para_in[1], prompt_path))

done <- if (file.exists(labels_path))
  readr::read_csv(labels_path, show_col_types = FALSE, col_types = "cc") else
  tibble(para_uid = character(), label = character())
todo <- dplyr::anti_join(sampled, done, by = "para_uid")
windows <- split(todo, ceiling(seq_len(nrow(todo)) / st>window_n))
type <- ellmer::type_object(label = ellmer::type_enum(
  c("favor", "neutral", "oppose", "irrelevant"), "The single best label."))

done <- purrr::reduce(seq_along(windows), function(done, i) {
  if (i > 1) purrr::walk(seq_len(st>window_wait), function(s) Sys.sleep(1),
    .progress = glue::glue("resting {st>window_wait}s"))
  d <- windows[[i]]
  message(glue::glue("window {i} of {length(windows)}: {nrow(d)} calls"))
  res <- ask_many(make_chat("stance", sys, st>steps, st>use_openrouter),
    str_c("Text: ", d$text), type, rpm = st>window_n)
  if (all(is.na(res$label)))
    stop("Every call in the window failed. First error: ",
      conditionMessage(purrr::compact(res$error)[[1]]))
  out <- dplyr::bind_rows(done, tibble(para_uid = d$para_uid,
    label = as.character(res$label))) |>

    dplyr::filter(!is.na(label))
  readr::write_csv(out, labels_path)
  record_inputs(labels_path, c(para_in[1], prompt_path))
  out
}, .init = done)

coded <- dplyr::inner_join(sampled, done, by = "para_uid")
if (nrow(coded) < nrow(sampled))
  writeLines(glue::glue("**{nrow(sampled)} - {nrow(coded)} calls failed;** re-
render to retry them."))
coded |>
  dplyr::count(stratum, label) |>
  tidyr::pivot_wider(names_from = label, values_from = n, values_fill = 0) |>
  dplyr::left_join(coded |>
    dplyr::group_by(stratum) |>
    dplyr::summarise(relevant_share = round(mean(label !=
"irrelevant"), 2),
    weight = round(dplyr::first(w), 1)),
    by = "stratum") |>
  dplyr::arrange(match(stratum, c("high", "mid", "low"))) |>
  tbl_paged(caption = "Labels by stratum (counts). relevant_share in the low
stratum is the audit: how often the similarity screen ranked a relevant paragraph
low. Each of those stands for `weight` paragraphs, so a high value there widens
every interval.")

```

Labels by stratum (counts). relevant_share in the low stratum is the audit: how often the similarity screen ranked a relevant paragraph low. Each of those stands for weight paragraphs, so a high value there widens every interval.

stratum	favor	irrelevant	neutral	oppose	relevant_share	weight
high	20	9	6	13	0.81	1.0
mid	75	81	17	17	0.57	1.0
low	4	58	0	0	0.06	11.5

What the labels look like

```

coded |>
  dplyr::group_by(label) |>
  dplyr::slice_max(sim, n = 3, with_ties = FALSE) |>
  dplyr::ungroup() |>
  dplyr::mutate(title = str_c(toupper(label), " | ", Year, " | ", para_uid),
                text = str_trunc(text, 350)) |>
  tbl_records("title", "text",
              caption = "The three paragraphs most similar to the claim for each
label. If these do not match what the label should mean, fix the claim before
reading any number.")

```

The three paragraphs most similar to the claim for each label. If these do not match what the label should mean, fix the claim before reading any number.

FAVOR | 1973 | 1973-Nixon-04 - text: The peace we seek in the world is not the flimsy peace which is merely an interlude between wars, but a peace which can endure for generations to come. It is important that we understand both the necessity and the limitations of America's role in maintaining that peace. Unless we in America work to preserve the peace, there will be no peace. Unl...

FAVOR | 2001 | 2001-Bush-09 - text: We will build our defenses beyond challenge, lest weakness invite challenge. We will confront weapons of mass destruction, so that a new century is spared new horrors. The enemies of liberty and our country should make no mistake: America remains engaged in the world by history and by choice, shaping a balance of power that favors freedom. We wi...

FAVOR | 1957 | 1957-Eisenhower-13 - text: We recognize and accept our own deep involvement in the destiny of men everywhere. We are accordingly pledged to honor, and to strive to fortify, the authority of the United Nations. For in that body rests the best hope of our age for the assertion of that law by which all nations may live in dignity. And, beyond this general resolve, we are cal...

IRRELEVANT | 2025 | 2025-Trump-27 - text: Above all, my message to Americans today is that it is time for us to once again act with courage, vigor, and the vitality of history's greatest

civilization. So, as we liberate our nation, we will lead it to new heights of victory and success. We will not be deterred. Together, we will end the chronic disease epidemic and keep our children safe...

IRRELEVANT | 2021 | 2021-Biden-24 - text: This is our historic moment of crisis and challenge, and unity is the path forward. And, we must meet this moment as the United States of America. If we do that, I guarantee you, we will not fail. We have never, ever, ever failed in America when we have acted together. And so today, at this time and in this place, let us start afresh. All of us....

IRRELEVANT | 2017 | 2017-Trump-19 - text: Finally, we must think big and dream even bigger. In America, we understand that a nation is only living as long as it is striving. We will no longer accept politicians who are all talk and no action - constantly complaining but never doing anything about it. The time for empty talk is over. Now arrives the hour of action. Do not let anyone tell...

NEUTRAL | 1909 | 1909-Taft-16 - text: Our international policy is always to promote peace. We shall enter into any war with a full consciousness of the awful consequences that it always entails, whether successful or not, and we, of course, shall make every effort consistent with national honor and the highest national interest to avoid a resort to arms. We favor every instrumentali...

NEUTRAL | 1973 | 1973-Nixon-10 - text: Just as building a structure of peace abroad has required turning away from old policies that failed, so building a new era of progress at home requires turning away from old policies that have failed. Abroad, the shift from old policies to new has not been a retreat from our responsibilities, but a better way to peace. And at home, the shift fr...

NEUTRAL | 1933 | 1933-Roosevelt-11 - text: The basic thought that guides these specific means of national recovery is not narrowly nationalistic. It is the insistence, as a first consideration, upon the interdependence of the various elements in all parts of the United States - a recognition of the old and permanently important manifestation of the American spirit of the pioneer. It is t...

OPPOSE | 1973 | 1973-Nixon-05 - text: We shall continue, in this era of negotiation, to work for the limitation of nuclear arms, and to reduce the danger of confrontation between the great powers. We shall do our share in defending peace and freedom in the world. But we shall expect others to do their share. The time has passed when America will make every other nation's conflict ou...

OPPOSE | 2017 | 2017-Trump-17 - text: We will follow two simple rules: buy American and hire American. We will seek friendship and goodwill with the nations of the world - but we do so with the understanding that it is the right of all nations to put their own interests first. We do not seek to impose our way of life on anyone, but rather to let it shine as an example for everyone t...

OPPOSE | 1849 | 1849-Taylor-06 - text: As American freemen we can not but sympathize in all efforts to extend the blessings of civil and political liberty, but at the same time we are warned by the admonitions of history and the voice of our own beloved Washington to abstain from entangling alliances with foreign nations. In all disputes between conflicting governments it is our inte...

Results

All shares are weighted to the whole corpus, with 95 percent intervals clustered by comment.

```
rel <- dplyr::filter(coded, label != "irrelevant")
sided <- dplyr::filter(rel, label %in% c("favor", "oppose"))
dplyr::bind_rows(
  dplyr::mutate(share_ci(coded$label != "irrelevant", coded$atom_id, coded$w),
    measure = "paragraphs addressing the claim", coded = nrow(coded)),
  dplyr::mutate(share_ci(rel$label == "favor", rel$atom_id, rel$w),
    measure = "favor, of those addressing it", coded = nrow(rel)),
  dplyr::mutate(share_ci(rel$label == "neutral", rel$atom_id, rel$w),
    measure = "neutral, of those addressing it", coded = nrow(rel)),
  dplyr::mutate(share_ci(sided$label == "favor", sided$atom_id, sided$w),
    measure = "favor, of those taking a side", coded = nrow(sided)))
|>
dplyr::select(measure, coded, share, lo, hi) |>
tbl_paged(caption = "Weighted shares for the whole corpus.")
```

Weighted shares for the whole corpus.

measure	coded	share	lo	hi
paragraphs addressing the claim	300	0.204	0.121	0.287
favor, of those addressing it	152	0.727	0.612	0.842
neutral, of those addressing it	152	0.119	0.061	0.176
favor, of those taking a side	129	0.825	0.730	0.919

Two lines over time. **Stance** is the share of paragraphs addressing the claim that favor it. **Attention** is the share of all paragraphs that address the claim at all. Read together, they separate a change in opinion from a change in how often the claim comes up.

```
from <- st$min_year %||% min(coded$Year)
fit_one <- function(d, y, what) {
  d <- dplyr::filter(d, Year >= from)
  r <- tryCatch(trend_curve(y(d), d$Year, d$atom_id, st$df_spline,
    st$min_class_n, d$w),
    error = function(e) conditionMessage(e))
  if (is.character(r)) {
    writeLines(glue::glue("**{what}: no trend fitted:** {r}."))
    return(NULL)
  }
  if (r$df[1] == 1L) writeLines(glue::glue("{what}: fewer than 10 paragraphs on
one side, so a straight line in year."))
  dplyr::mutate(r, line = what)
}
rug <- dplyr::bind_rows(
```

```

dplyr::transmute(dplyr::filter(rel, Year >= from), Year,
  line = "stance: favor, of those addressing the claim"),
dplyr::transmute(dplyr::filter(coded, Year >= from), Year,
  line = "attention: share addressing the claim"))
trends <- dplyr::bind_rows(
  fit_one(rel, function(d) as.integer(d$label == "favor"),
    "stance: favor, of those addressing the claim"),
  fit_one(coded, function(d) as.integer(d$label != "irrelevant"),
    "attention: share addressing the claim"))
if (nrow(trends)) readr::write_csv(trends, p_out(str_c("stance_trend_",
st$name, ".csv")))

```

```

if (nrow(trends))
  ggplot(trends, aes(Year, fit)) +
    geom_ribbon(aes(ymin = lo, ymax = hi), alpha = 0.2, fill = viridis(1, begin
= 0.35)) +
    geom_line(linewidth = 0.9, color = viridis(1, begin = 0.35)) +
    geom_rug(data = dplyr::semi_join(rug, trends, by = "line"), aes(Year),
      inherit.aes = FALSE, alpha = 0.4, length = grid::unit(0.03, "npc"))
+
  facet_wrap(~ line, ncol = 1, scales = "free_y") +
  scale_y_continuous(labels = function(x) str_c(round(100 * x), "%")) +
  labs(x = NULL, y = NULL)

```

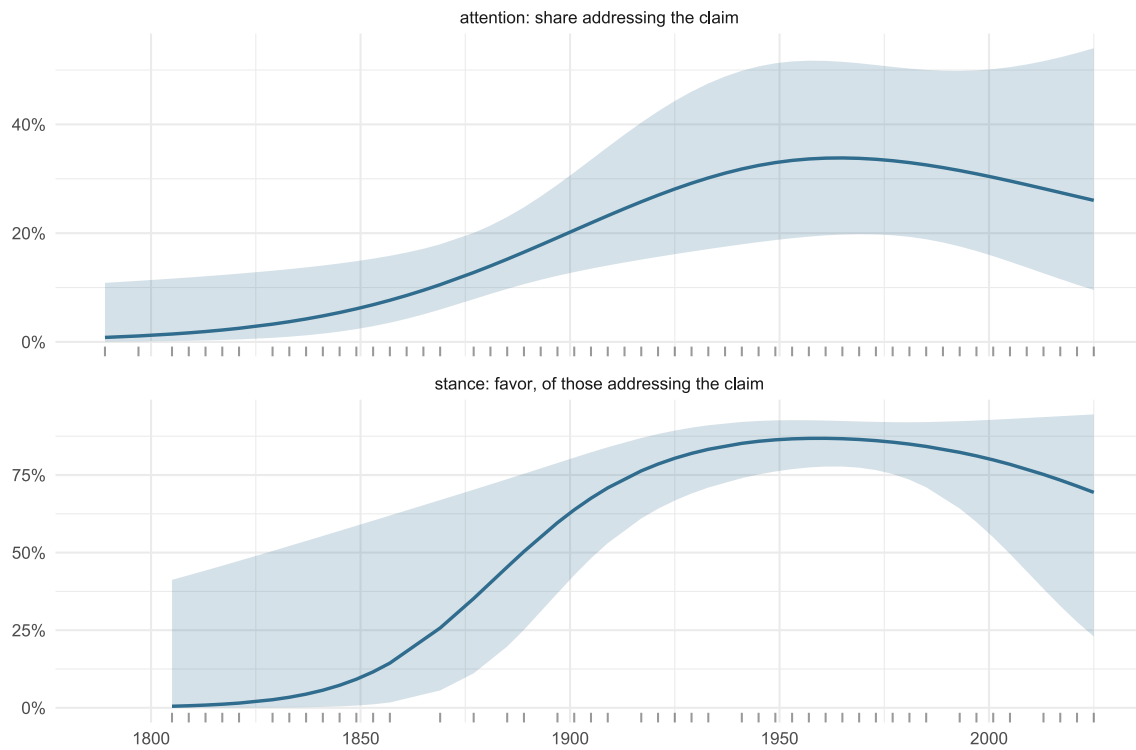


Figure 1: Weighted trends with 95 percent intervals clustered by comment. Ticks along the bottom mark the coded paragraphs behind each line; where they are sparse, the line is mostly the model's smoothing. The intervals do not include error in the model's labels.

Before using these numbers

This step is experimental. Labels come from one model with no human check, so their error is not in the intervals. Relevant paragraphs found in the low stratum carry large weights and widen the intervals; if many turn up there, move budget from the high stratum to the middle. Every share is a share of paragraphs, so a long comment counts more than a short one.